

JUNCHENG LU

jl7191@tc.columbia.edu | 773-328-0805 |  Personal Website |  GitHub

EDUCATION

Teachers College, Columbia University

September 2025 – Present

M.A. in Cognitive Science in Education

GPA: 3.90 / 4.00

Beijing Normal University

September 2020 – July 2024

B.S. in Psychology

Selected coursework: Computational Cognitive Neuroscience; Brain Imaging Data Modeling;
fMRI: Principles, Experiment Design and Data Analysis.

RESEARCH EXPERIENCE

Social Cognitive and Neural Sciences Lab

September 2025 – Present

Research Assistant, Columbia University | PI: Dr. Jon Freeman

Metacognitive Accuracy in Implicit Bias Prediction

First author; manuscript in preparation

January 2026 – Present

- Investigate whether people predict ingroup-affirming implicit biases more accurately than outgroup-affirming biases, using 2,986 participant-IAT observations from 1,866 participants across four race and age IATs.
- Develop R workflows linking MPT/Quad and drift-diffusion model parameters to directional and continuous prediction accuracy, accounting for explicit attitudes in mixed-effects models.
- Test the incremental explanatory value of process parameters through nested model comparisons and assess estimation reliability using convergence diagnostics; manuscript in preparation.

Computational Modeling of the Weapon Identification Task

Independent project

September – December 2025

- Applied DDM and MPT models to examine how evidence accumulation and latent processing components contribute to biased weapon-identification decisions.
- Compared competing cognitive architectures and related model-derived parameters to fMRI activation to evaluate accounts of threat processing and cognitive control.

Intergroup Relation Neuroscience Lab

March 2022 – June 2025

Research Assistant, Beijing Normal University | PI: Dr. Xiaochun Han

Computational and Neural Mechanisms of Collateral-Harm Decisions

R&R at Journal of Experimental Social Psychology; co-first author

March 2022 – June 2025

- Led study design, experimental implementation, and data collection in collaboration with the PI across two studies (total $N = 983$), combining a cross-cultural behavioral experiment with EEG.
- Conducted all behavioral, computational, and EEG analyses, using hierarchical Bayesian DDMs and P2/N2 measures to examine how civilian race and pain expressions relate to decision caution.
- Identified a Race $\times$ Expression interaction in decision caution despite no corresponding condition differences in overt attack decisions; contributed to manuscript drafting and revision with the PI.

Social Roles and Civilian–Soldier Judgments in Intergroup Conflict

Co-first author; manuscript in preparation

September 2023 – June 2025

- Conducted data analyses for four studies examining social-role information, civilian–soldier differences in attack decisions and bystander judgments, and representations of warmth and competence.
- Fit logistic and linear mixed-effects models accounting for participants and social roles; tested cross-cultural patterns, category-by-information interactions, and experimental manipulations of warmth and competence.
- Analyses showed smaller civilian–soldier differences with social-role information across decision and representation tasks; contributed to manuscript writing.

MANUSCRIPTS

UNDER REVIEW

Lu, J.^{*}, Zhang, Y.^{*}, Han, X., Xiao, Z., & Han, X.[†] (2026). Computational and neural mechanisms of race-dependent influence of civilian suffering on collateral-harm decisions during intergroup conflicts. R&R at *Journal of Experimental Social Psychology*.

IN PREPARATION

Zhang, Y.^{*}, Lu, J.^{*}, & Han, X.[†] (2026). Cognitive mechanism of killing-decision discrimination between combatants and civilians during intergroup conflict: The influence of social-role information. *Manuscript in preparation*.

^{*} Equal contribution. [†] Corresponding author.

CONFERENCE

Lu, J., & Han, X.[†] (April 2024). Computational and neural mechanisms of decision-making causing same-race vs. other-race civilian casualties during intergroup conflicts. Accepted for oral and poster presentation at the *Social and Affective Neuroscience Society Annual Meeting*, Toronto, Canada. Unable to present due to visa restrictions.

TECHNICAL SKILLS

Programming	Python (HDDM, NumPy, pandas); R (brms, lme4, TreeBUGS, rtmpt); MATLAB (EEGLAB, SPM12, Psychtoolbox); JavaScript/HTML (jsPsych).
Modeling & Statistics	Hierarchical Bayesian modeling; DDM and MPT/Quad models; GLMM/LMM; MCMC diagnostics; model comparison.
Neuroimaging	EEG/ERP preprocessing and analysis; fMRI preprocessing and modeling; model-based brain-behavior analysis.